ZHI WANG **Zhongguancun Laboratory** **Assistant Researcher**

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SUMMARY

I study how emerging technologies, including neural networks, large language models (LLMs), blockchains, and online social networks, can be abused by attackers, and how to detect and defend against these new threats. My recent work covers LLM security (LLM-powered honeypots and package hallucinations in software supply chains), AI-enabled malware, and covert command-and-control. I welcome collaboration on LLM and AI security, software supply chain security, and AI-driven threat detection.

EMPLOYMENT

2023 – Present **Zhongguancun Laboratory**, Assistant Researcher

EDUCATION

2017 – 2023 **UCAS / Institute of Information Engineering, CAS**, *Cyber Security*, Ph.D.

2013 – 2017 **Civil Aviation University of China**, *Information Security*, B.Eng.

RESEARCH INTERESTS

AI & LLM Security LLM security, backdoor and adversarial attacks on ML

Malware & Botnets AI-enabled malware, covert command-and-control

RESEARCH PROJECTS

National Natural Science Foundation of China (NSFC)	Principal Investigator	2027 – 2029
National Science and Technology Major Project	Sub-project Leader	2025 – 2028
National Laboratory Special Program	Key Member	2023 – 2025

PUBLICATIONS & PATENTS

Selected Papers Full list on Google Scholar

- Zhe Huang, **Zhi Wang***, Chaoge Liu, Xiao Deng, Shengjia Chang, Mengyang Xu. Beyond the Blank Slate: Typed Operating System State for Cross-Session Consistency in LLM-Powered Honeypots. In 20th International Symposium on Research in Attacks, Intrusions and Defenses (**RAID**), Lancaster, UK. Springer, 2026. DOI: 10.1007/978-3-032-42929-2_20
- Zhe Huang, Chaoge Liu*, **Zhi Wang**, Xiang Cui, Xiao Deng, Mengyang Xu. Package Hallucinations as Phantoms in Open-source Software Supply Chains: An Empirical Security Analysis. **ACM Transactions on Software Engineering and Methodology (TOSEM)**. 2026, 35(11): Article 363. DOI: 10.1145/3830237
- Yong Zhao, Ang Xia, Jie Yin, **Zhi Wang**, Yaqin Cao*, Xiangyi Zeng, Yuling Liu. BASSET: Enhancing Binary Code Clone Searching through Multi-Level Hybrid Semantic Indexing. In 2025 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (**DSN**), Naples, Italy. IEEE, 2025: 47-60. DOI: 10.1109/DSN64029.2025.00020
- Chaoge Liu, **Zhi Wang***, Jie Yin, Yinsheng Liu, Kai Mao, Zhe Huang, Chumeng Deng. SCBot: Building Lightweight and Flexible C&C Based on Smart Contract. In 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**), Hyderabad, India. IEEE, 2025: 1-5. DOI: 10.1109/ICASSP49660.2025.10889707

- Xutong Wang, Chaoge Liu*, Xiaohui Hu, **Zhi Wang**, Jie Yin, and Xiang Cui. Make Data Reliable: An Explanation-powered Cleaning on Malware Dataset Against Backdoor Poisoning Attacks. In Annual Computer Security Applications Conference (**ACSAC**), ACM, 2022: 267 – 278. DOI: 10.1145/3564625.3564661
- **Zhi Wang**, Chaoge Liu*, Xiang Cui*, Jie Yin, Jiaxi Liu, Di Wu, and Qixu Liu. DeepC2: AI-Powered Covert Command and Control on OSNs. In 24th International Conference on Information and Communications Security (**ICICS**). Springer, Cham, 2022: 394-414. DOI: 10.1007/978-3-031-15777-6_22 (**Best Artifact Award**)
- **Zhi Wang**, Chaoge Liu*, Xiang Cui*, Jie Yin, and Xutong Wang. EvilModel 2.0: Bringing Neural Network Models into Malware Attacks. **Computers & Security**. 2022, 120: 102807. DOI: 10.1016/j.cose.2022.102807
- **Zhi Wang**, Chaoge Liu*, and Xiang Cui. EvilModel: Hiding Malware Inside of Neural Network Models. In 2021 IEEE Symposium on Computers and Communications (**ISCC**). IEEE, 2021: 1-7. DOI: 10.1109/iscc53001.2021.9631425

Patents

- Qixu Liu, Jianjun Zhao, Ru Tan, **Zhi Wang**, Feng Dai, Yaqin Cao, Xingchen Chen. A Threat Intelligence Sharing Method and Apparatus Compliant with Data Security Requirements. Chinese Invention Patent, ZL202210010348.3.

</> Open-Source Artifacts

- **EvilModel**: hiding malware inside neural network models — github.com/oicid/EvilModel
- **DeepC2**: AI-powered covert C&C on online social networks — github.com/oicid/DeepC2

Awards & Honors

Competitions & Awards

ICICS 2022	*Best Artifact Award*	Sep. 2022
DataCon, Network Traffic Analysis Track	*First Prize*	Oct. 2021
Coremail E-mail Security Analysis Contest	*Excellence Award*	Oct. 2020
National College Student Information Security Contest	*First Prize*	Aug. 2015

Professional Service

- **Reviewer** Journals: Transactions on Information Forensics & Security, Computers & Security, Cybersecurity, Blockchain: Research and Applications, Blockchain, Journal of Cyber Security. Conferences: IJCNN 2025–2026, ICASSP 2025
- **TPC Member** TheWebConf (WWW) 2026, ICAIDES 2025–2026
- **Membership** IEEE, ACM; China Computer Federation (CCF), Executive Member of the CCF Technical Committees on Network and System Security and on Information Security

(Last updated: 2026-11-06)